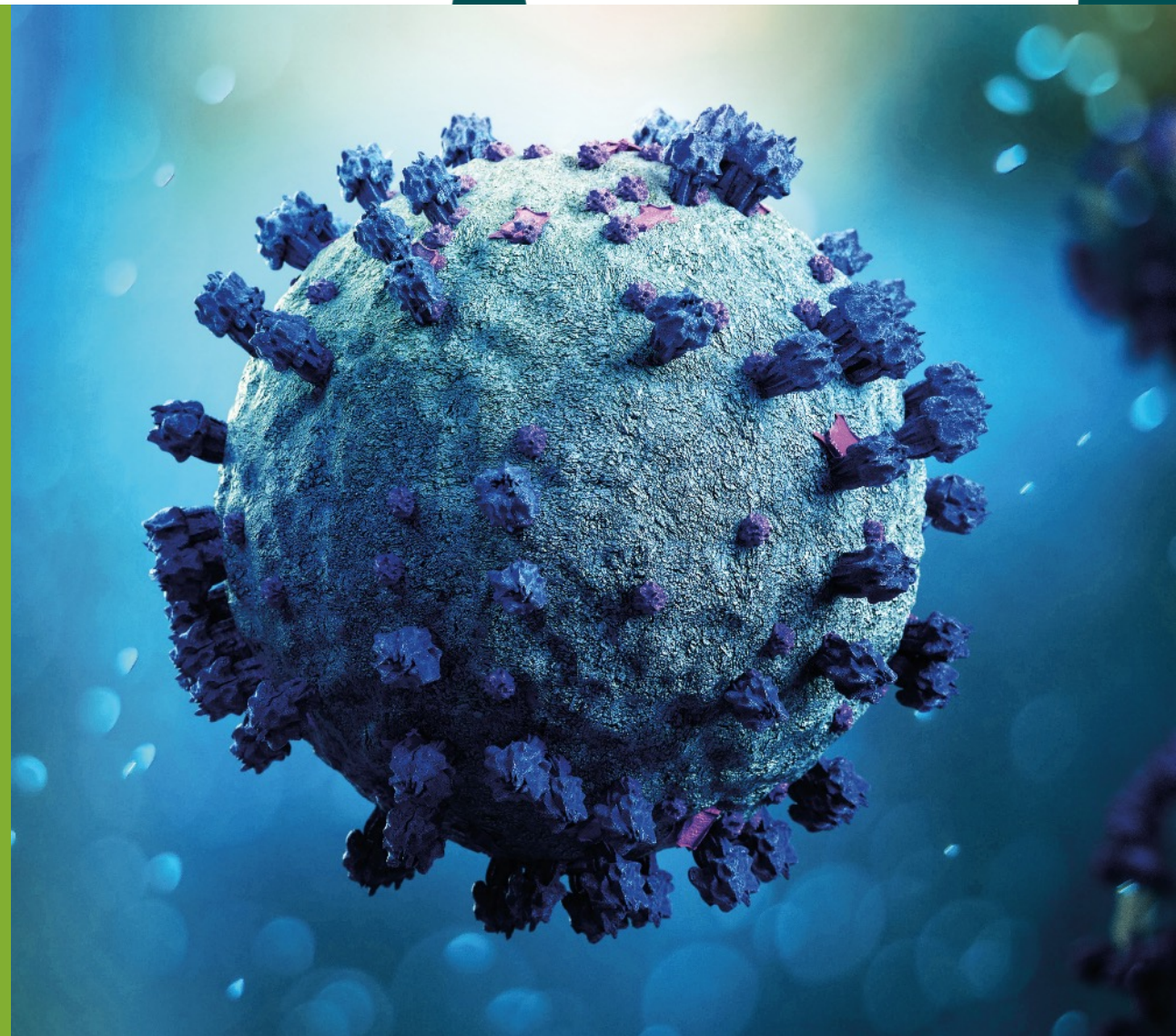


CHAPTER 10

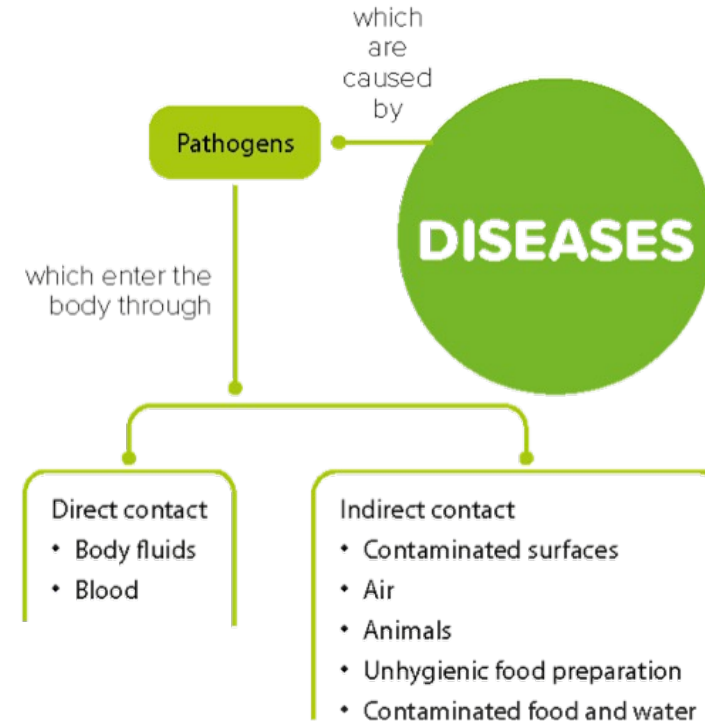
DISEASES, IMMUNITY AND DRUGS



10.1 What Is a Disease?

In this section, you will learn the following:

- Describe a pathogen as a disease-causing organism.
- Describe a transmissible disease as a disease in which the pathogen can be passed from one host to another.
- Describe the body defences.
- State that a pathogen is transmitted through direct and indirect contact.
- Explain the importance of various factors in controlling the spread of diseases.
- **S** Describe cholera as a disease caused by a bacterium that is transmitted in contaminated water.
- **S** Explain how the cholera bacterium can cause diarrhoea, dehydration and loss of ions from the blood.



- 1 Describe a pathogen as a disease-causing organism
- 2 Describe a transmissible disease as a disease in which the pathogen can be passed from one host to another
- 3 State that a pathogen is transmitted:
 - (a) by direct contact, including through blood and other body fluids
 - (b) indirectly, including from contaminated surfaces, food, animals and air
- 4 Describe the body defences, limited to: skin, hairs in the nose, mucus, stomach acid and white blood cells
- 5 Explain the importance of the following in controlling the spread of disease:
 - (a) a clean water supply
 - (b) hygienic food preparation
 - (c) good personal hygiene
 - (d) waste disposal
 - (e) sewage treatment (details of the stages of sewage treatment are **not** required)

Pathogens

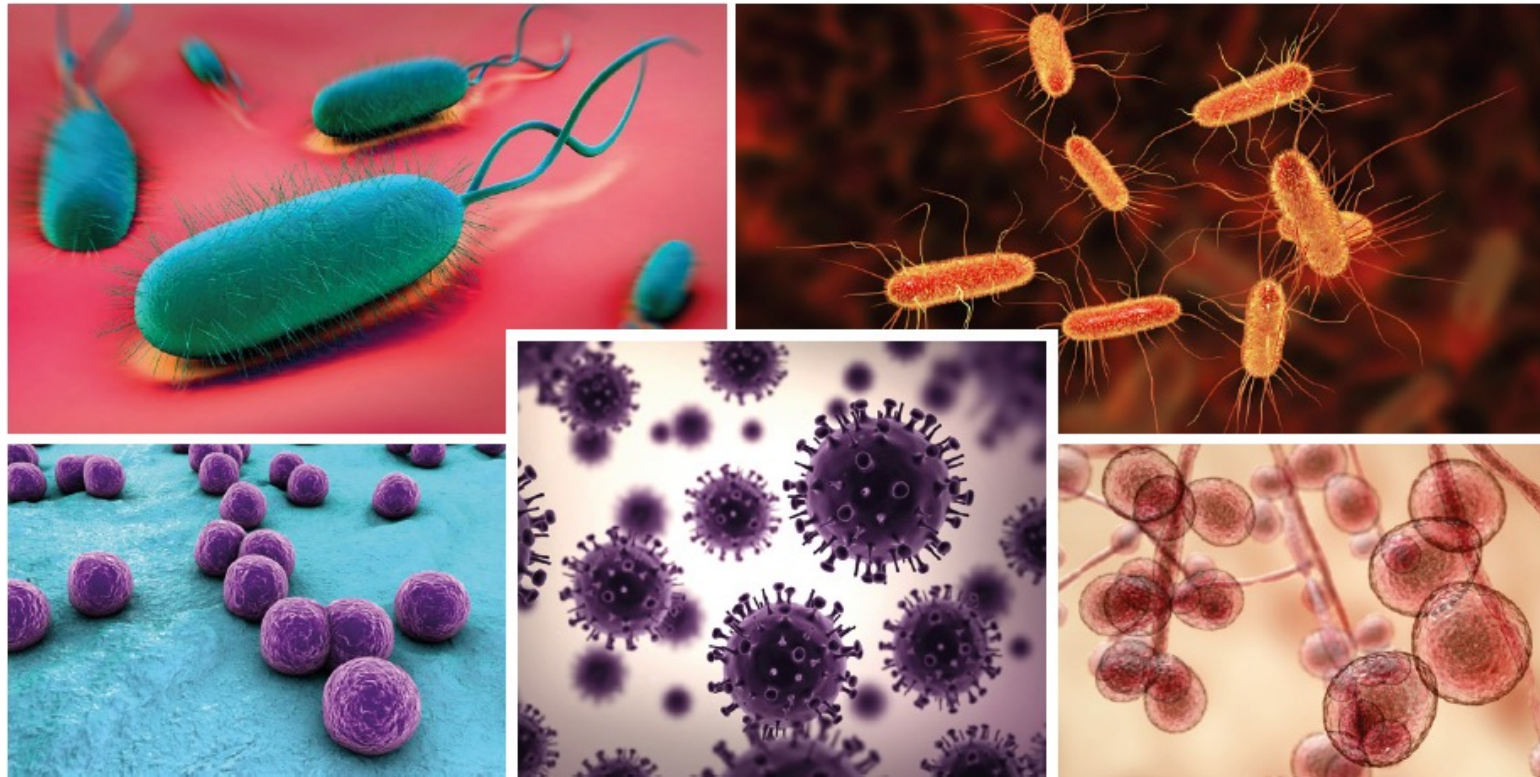
- A pathogen is a **disease-causing organism**
- Pathogens are passed on from one host to another and therefore the diseases they cause are known as **transmissible diseases**
- Pathogens can be passed on from host to host in different ways, including:
 - **Direct contact** - the pathogen is passed directly from one host to another by transfer of body fluids such as blood or semen (eg HIV, gonorrhoea, hepatitis B & C)
 - **Indirect contact** - the pathogen leaves the host and is carried in some way to another, uninfected individual

METHOD OF TRANSMISSION	EXAMPLES OF DISEASES SPREAD IN THIS WAY
DROPLETS IN AIR	COMMON COLD, INFLUENZA
FOOD OR WATER	CHOLERA, TYPHOID, DYSENTERY
TOUCHING CONTAMINATED SURFACES	ATHLETES FOOT, SALMONELLA (CAN BE TRANSMITTED ON THE FEET OF FLIES WHO LAND ON FOOD THAT IS THEN EATEN)
INSECT BITES	MALARIA, DENGUE FEVER

What are the organisms that cause diseases in humans?

Organisms that cause diseases are called **pathogens**.

Transmissible diseases are diseases in which the pathogen can be passed from one host to another.



Bacteria, viruses and fungi



Face Covering and Mask to Minimise Droplet Dispersion & Aerosolisation



Defences against Pathogens

There are **3** main ways in which the body defends itself against disease:**1. Mechanical barriers** – structures that make it difficult for pathogens to get past them and into the body

a) **Skin** - covers almost all parts of your body to prevent infection from pathogens. If it is cut or grazed, it immediately begins to heal itself, often by forming a scab.

b) **Hairs in the nose** - these make it difficult for pathogens to get past them further up the nose so they are not inhaled into the lungs

2. Chemical barriers – substances produced by the body cells that trap / kill pathogens before they can get further into the body and cause disease

a) **Mucus** - made in various places in the body, pathogens get trapped in the mucus and can then be removed from the body (by coughing, blowing the nose, swallowing etc)

b) **Stomach acid** - contains hydrochloric acid which is strong enough to kill any pathogens that have been caught in mucus in the airways and then swallowed or have been consumed in food or water

3. Cells - different types of white blood cell work to prevent pathogens reaching areas of the body they can replicate in

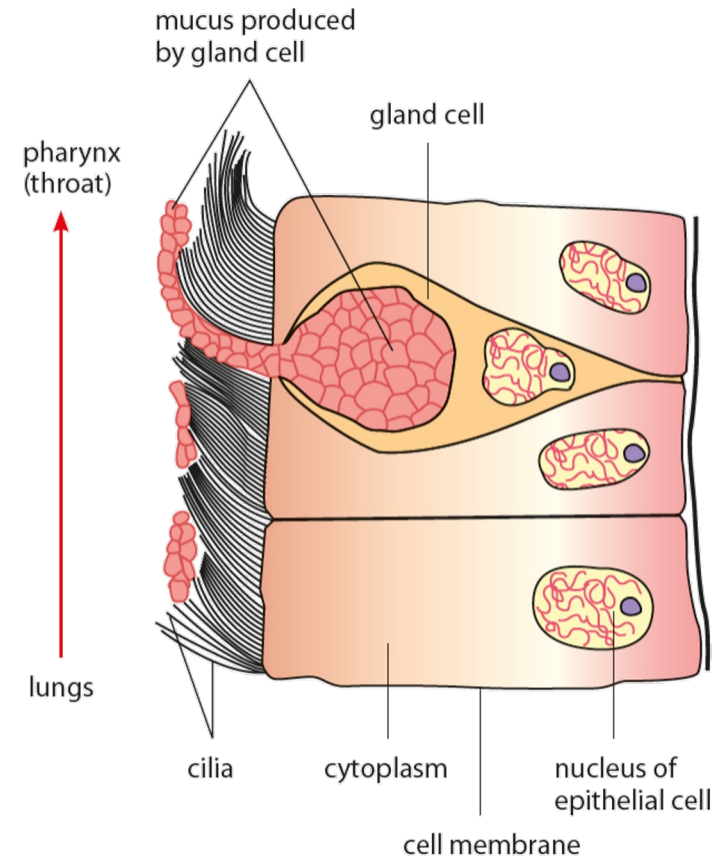
a) By **phagocytosis** - engulfing and digesting pathogenic cells

b) By producing **antibodies** - which clump pathogenic cells together so they can't move as easily (known as **agglutination**) and releasing chemicals that signal to other cells that they must be destroyed

External defence system

Our bodies have natural defence mechanisms to prevent pathogens from entering our bloodstream:

- Skin
- Hairs in the external nostrils
- Mucous membranes
- Stomach acid



Epithelial cells with cilia

How do pathogens get into our bodies?

Direct contact

- a) Person-to-person contact
- b) Exchange of body fluids
- c) Breastfeeding (mother to infant)



Indirect contact

- a) Contaminated surfaces
- b) Airborne infection
- c) Animals
- d) Unhygienic food preparation
- e) Contaminated food and water





How can the spread of waterborne and foodborne diseases be controlled?

(a) Having a clean water supply



(b) Having a proper waste disposal



(c) Having a proper sewage treatment



(d) Preparing food in a hygienic manner



(e) Practising good personal hygiene





- 16 Describe cholera as a disease caused by a bacterium which is transmitted in contaminated water
- 17 Explain that the cholera bacterium produces a toxin that causes secretion of chloride ions into the small intestine, causing osmotic movement of water into the gut, causing diarrhoea, dehydration and loss of ions from the blood

Cholera

Cholera is a disease caused by a bacterium that is transmitted through contaminated food or water.

Treatment of cholera

- Drink water containing some sugar and ions
- Intravenous fluid
- Antibiotics



Cholera bacteria



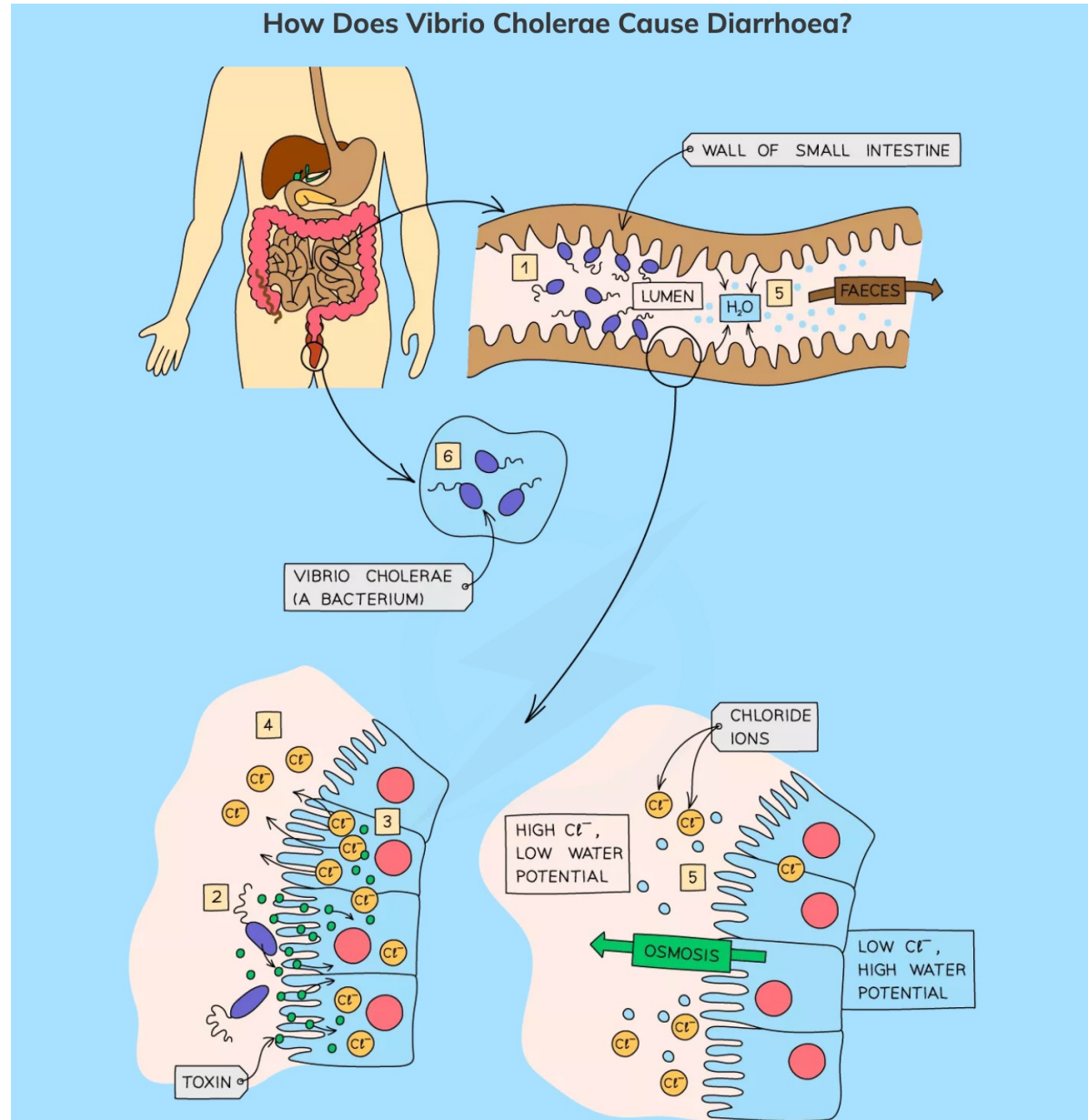
HELPFUL NOTES

Secretion of chloride ions out of the cells occurs by active transport. The cells use energy from respiration to pump chloride ions out of the cells against their concentration gradient.



Diarrhoea Causes & Treatment

- Diarrhoea is the **loss of watery faeces from the anus**
- If it is severe and continues for a long time, it **can lead to death**
- Severe diarrhoea can cause the **loss of significant amounts of water and ions** from the body, causing the tissues and organs to stop working properly
- It can be effectively treated by **oral rehydration therapy**
- This is a **drink with a small amount of salt and sugar** dissolved in it
- There are many causes of diarrhoea, one of which is infection with **Vibrio cholerae bacteria**, which causes the disease **cholera**



- Ingested via **infected water or food**, if it enters the small intestine it can cause illness in the following way:
 1. Bacteria attach to the wall of the **small intestine**
 2. They produce a **toxin**
 3. The toxin stimulates the cells lining the intestine to **release chloride ions** from inside the cells into the lumen of the intestine
 4. The chloride ions accumulate in the lumen of the small intestine and **lower the water potential** there
 5. Once the water potential is lower than that of the cells lining the intestine, **water starts to move out of the cells** into the intestine (by **osmosis**)
 6. Large quantities of water are lost from the body in **watery faeces**
 7. The blood contains **too little chloride ions and water**

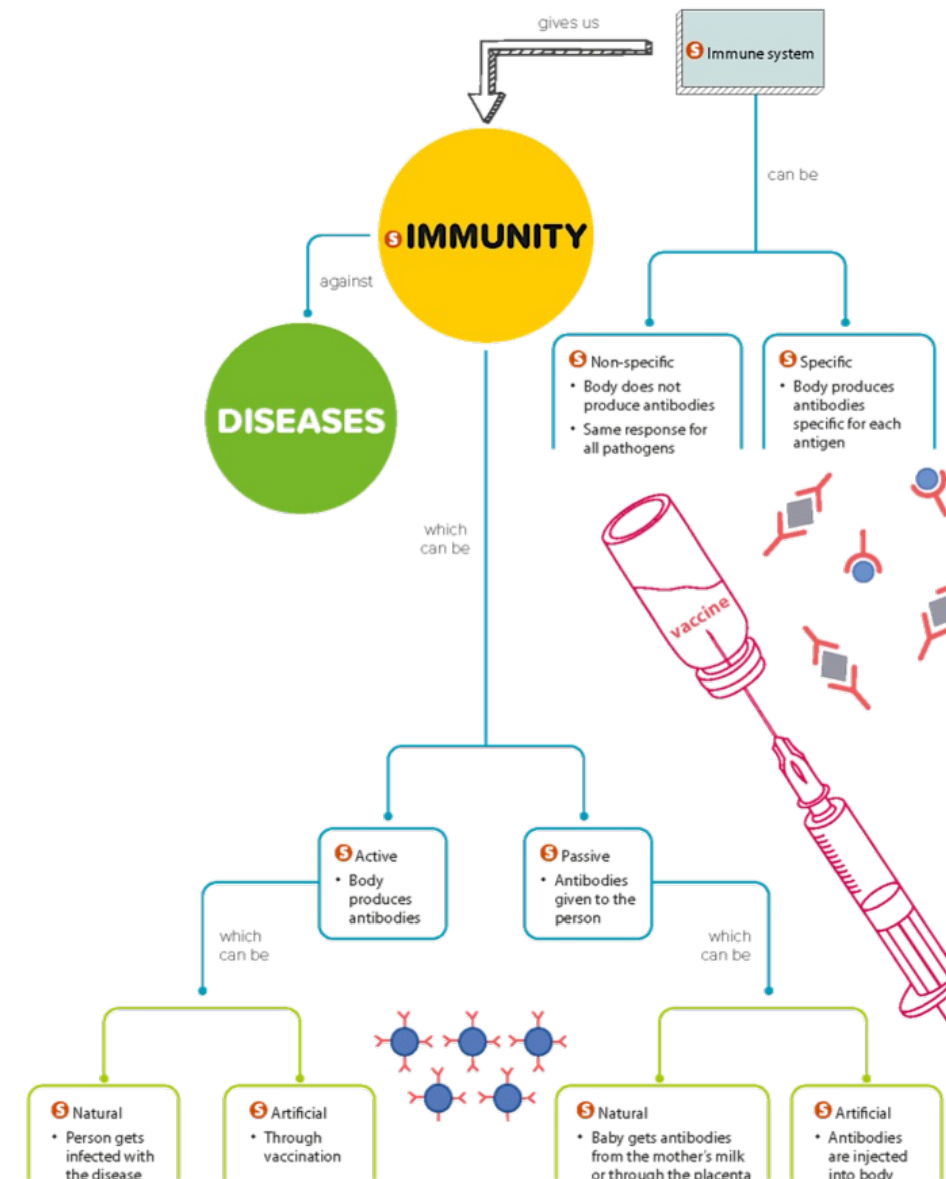
Let's Practise 10.1

- 1 What is a disease?
- 2 (a) What is a transmissible disease?
(b) Name a method by which each of these diseases is transmitted:
(i) AIDS (ii)  Cholera (iii) Common flu
- 3 State **three** methods by which diseases are spread.

S 10.2 Immunity

In this section, you will learn the following:

- State that each pathogen has its own antigens, which have specific shapes.
- Describe what antibodies are.
- State that specific antibodies have complementary shapes that fit specific antigens.
- Describe active immunity as defence against a pathogen by antibody production in the body.
- Explain that active immunity is gained after an infection by a pathogen or by vaccination.
- Explain that passive immunity is a short-term defence against a pathogen by antibodies acquired from another individual.
- Explain the importance of breastfeeding for the development of passive immunity in infants.
- Outline the process of vaccination.
- Explain the role of vaccination in controlling the spread of diseases.
- State that memory cells are not produced in passive immunity.



- 6 Describe active immunity as defence against a pathogen by antibody production in the body
- 7 State that each pathogen has its own antigens, which have specific shapes
- 8 Describe antibodies as proteins that bind to antigens leading to direct destruction of pathogens or marking of pathogens for destruction by phagocytes
- 9 State that specific antibodies have complementary shapes which fit specific antigens
- 13 Explain that passive immunity is a short-term defence against a pathogen by antibodies acquired from another individual, including across the placenta and in breast milk
- 14 Explain the importance of breast-feeding for the development of passive immunity in infants
- 15 State that memory cells are not produced in passive immunity

S

Immune system

Immunity is the protection provided by the body's defence system against a disease.

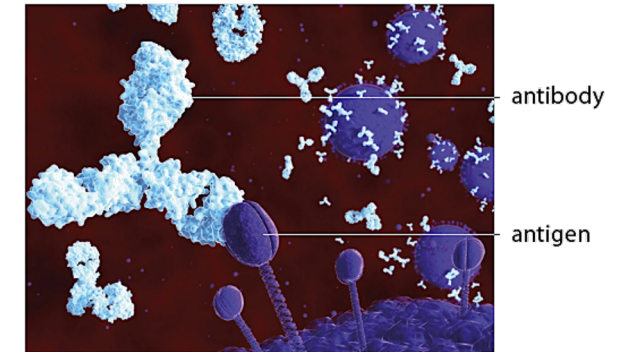
Specific immune system (or adaptive immune system)

When pathogens enter the bloodstream, they are destroyed by antibodies.

Antibodies are proteins produced by lymphocytes. They are highly specific.

Each pathogen has its own distinct antigen. Only an antibody of a specific shape complementary to the antigen can bind to the antigen, forming an antibody-antigen complex.

Antibodies play an important role in destroying foreign particles.



Antibody binding to an antigen



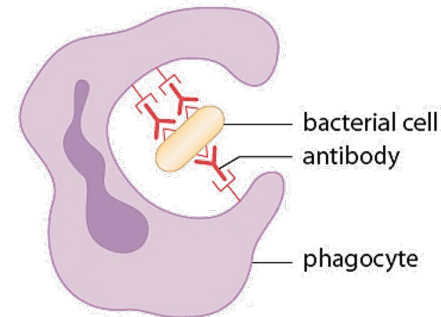
Antigens & Antibodies

- All cells have proteins and other substances projecting from their cell membrane
- These are known as **antigens** and are **specific** to that type of cell
- Lymphocytes have the ability to 'read' the antigens on the surfaces of cells and recognise any that are foreign
- They then make **antibodies** which are a **complementary shape to the antigens** on the surface of the pathogenic cell

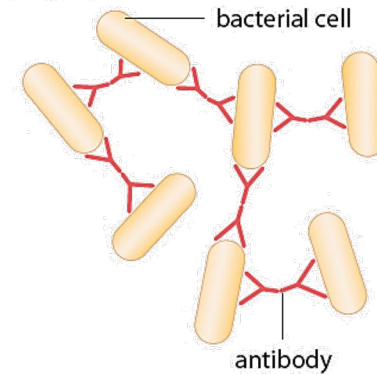
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Specific immune system (or adaptive immune system)

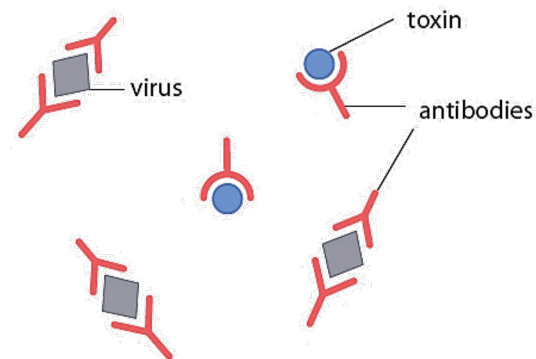
Antibodies bind to bacteria, marking them for phagocytes to engulf and destroy.



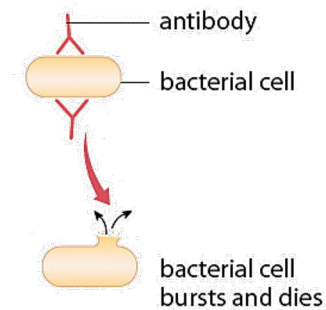
Antibodies bind to bacteria, causing them to clump together. This allows for easy ingestion by phagocytes.



Antibodies bind to toxins produced by bacteria, making them harmless. They also bind to viruses, making them inactive.



Antibodies bind to bacterial cells, causing the cells to burst and die.



Actions of antibodies



Active & Passive Immunity

Active immunity:

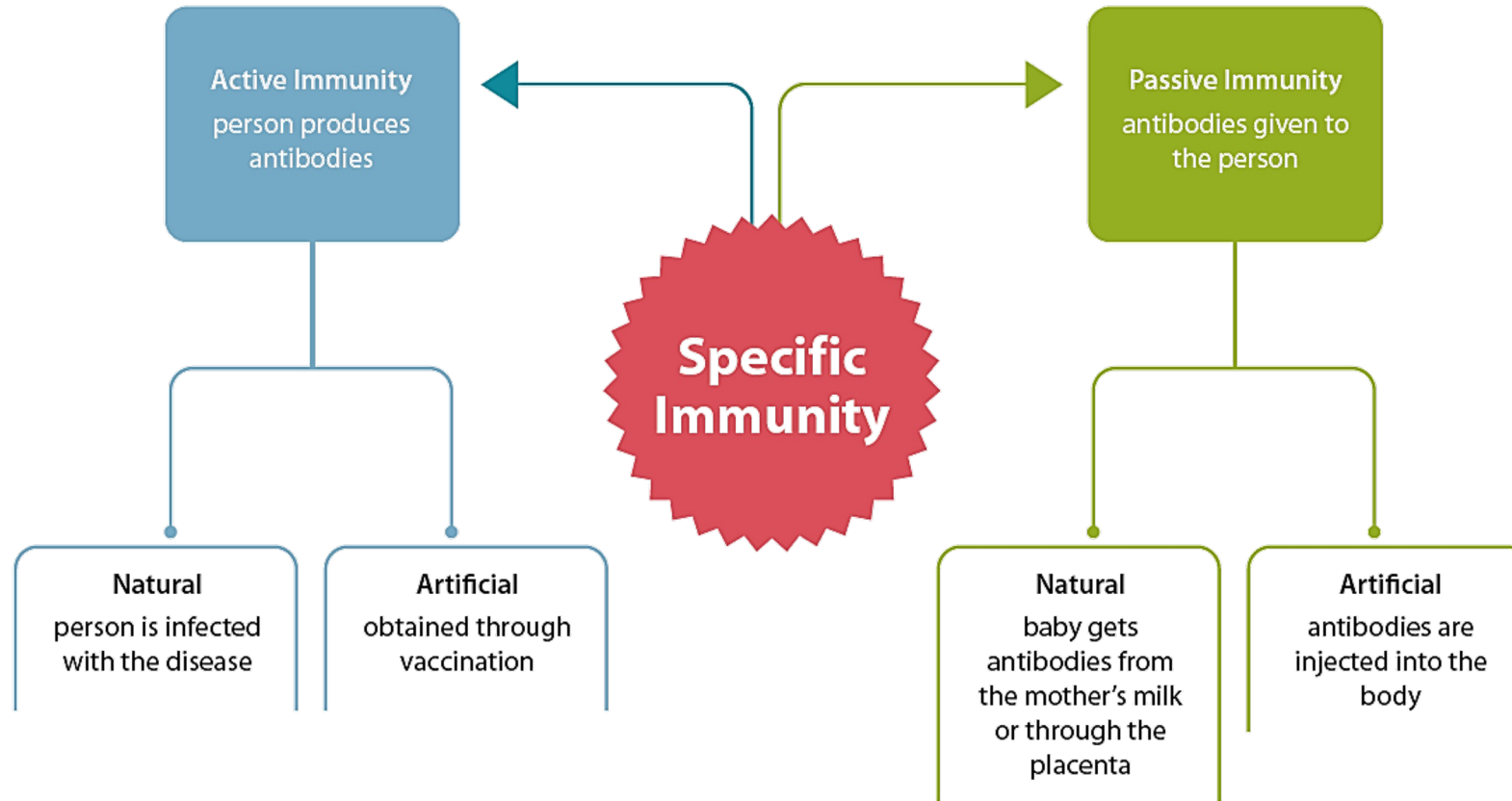
- Making antibodies and developing memory cells for future response to infection is known as **active immunity**
- There are **two ways** in which this active immune response happens:
 - The body has become **infected with a pathogen** and so the lymphocytes go through the process of making antibodies specific to that pathogen
 - **Vaccination**
- Active immunity is **slow acting** and provides **long-lasting immunity**

Passive immunity:

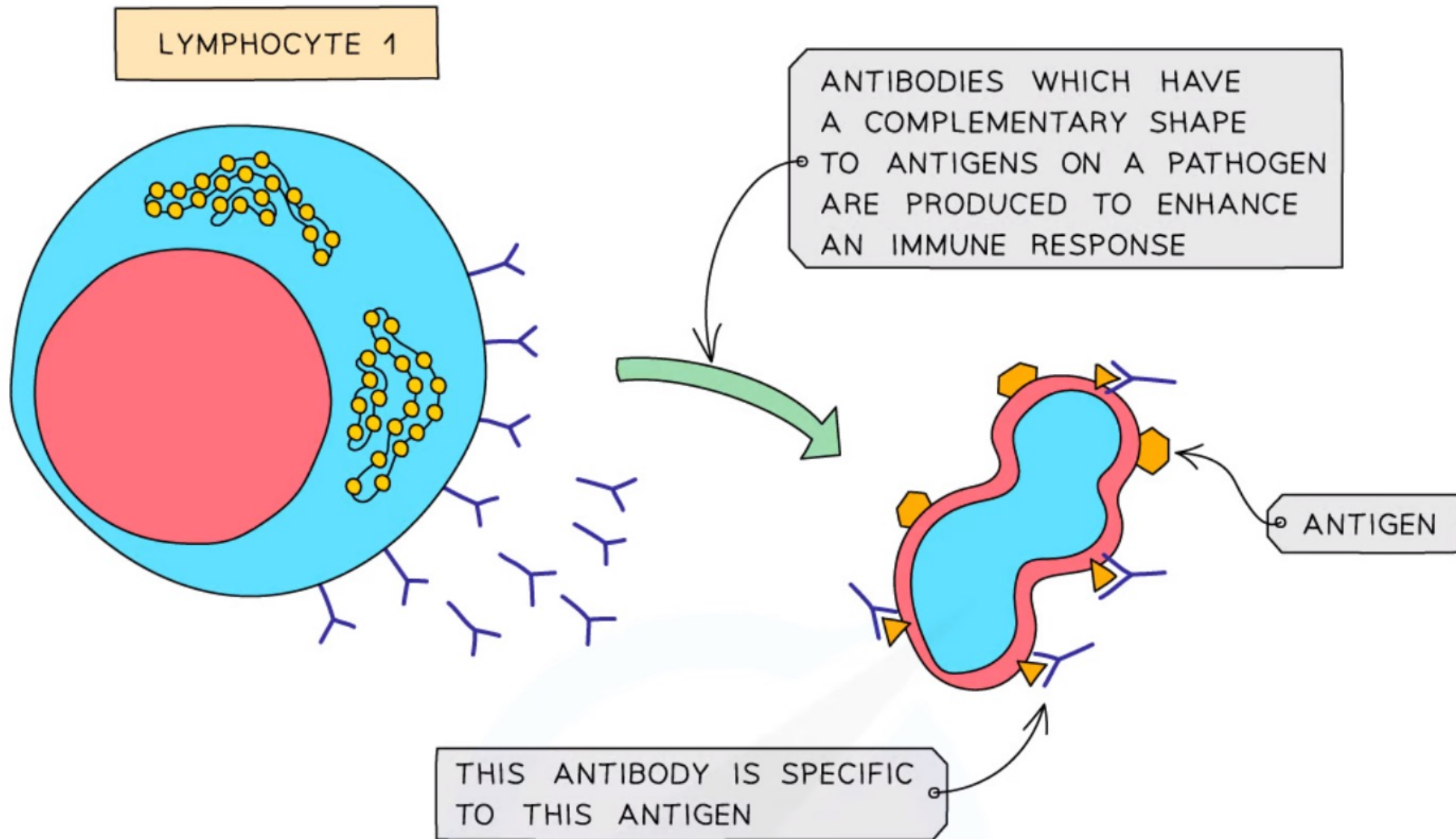
- This is when **ready-made antibodies**, from another source, are introduced to the body
- Passive immunity is a **fast-acting, short-term defence** against a pathogen by antibodies acquired from another individual, eg:
 - From **mother to infant** via **breast milk** - this is important as it helps the very young to fight off infections until they are **older and stronger** and their immune system is more responsive
 - **Injected antibodies** for certain diseases where the individual is already infected and a fast response is required, like **rabies** or **tetanus**
- The body **does not make its own antibodies or memory cells** in passive immunity, hence the name

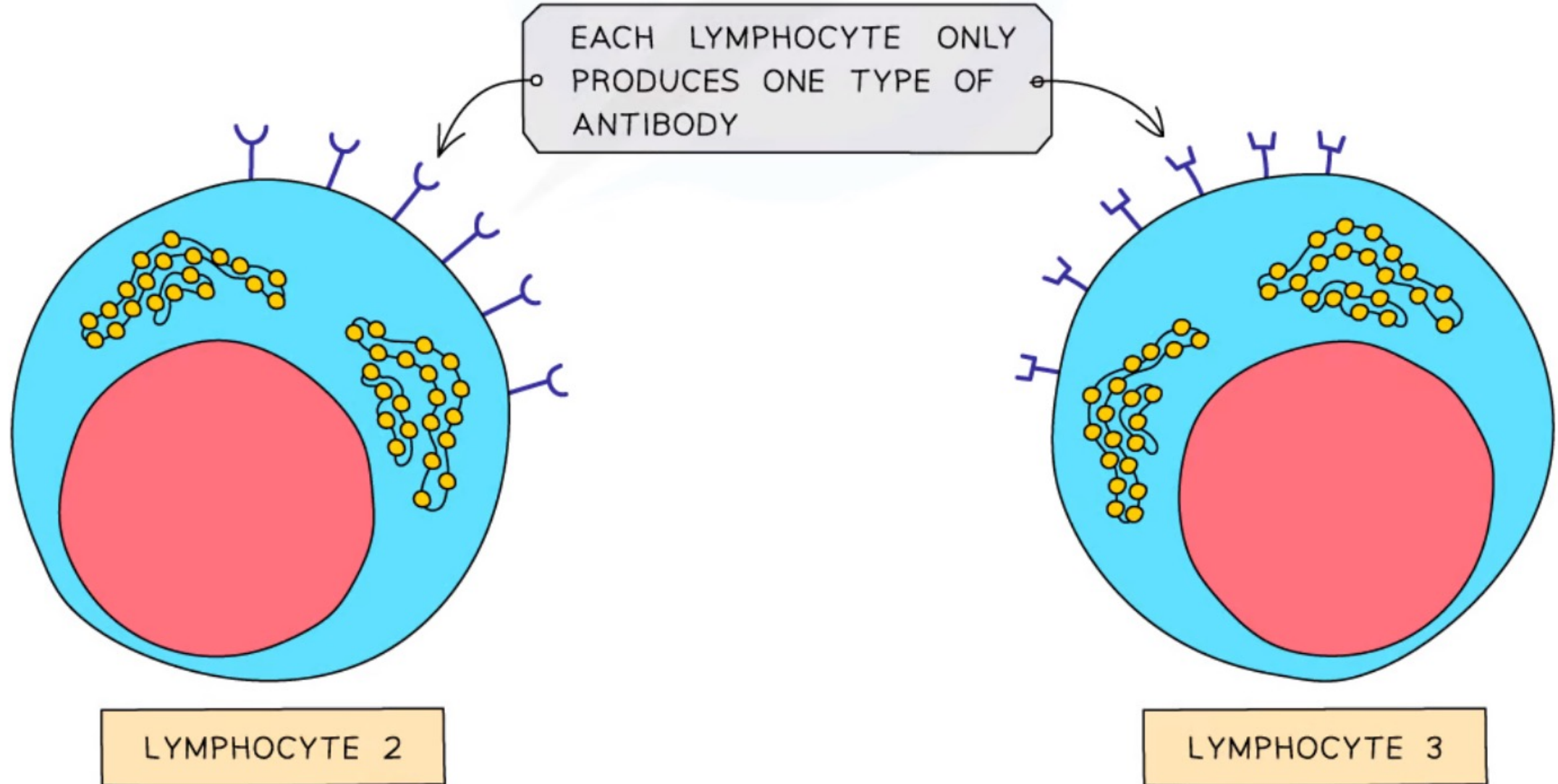
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Types of specific immunity

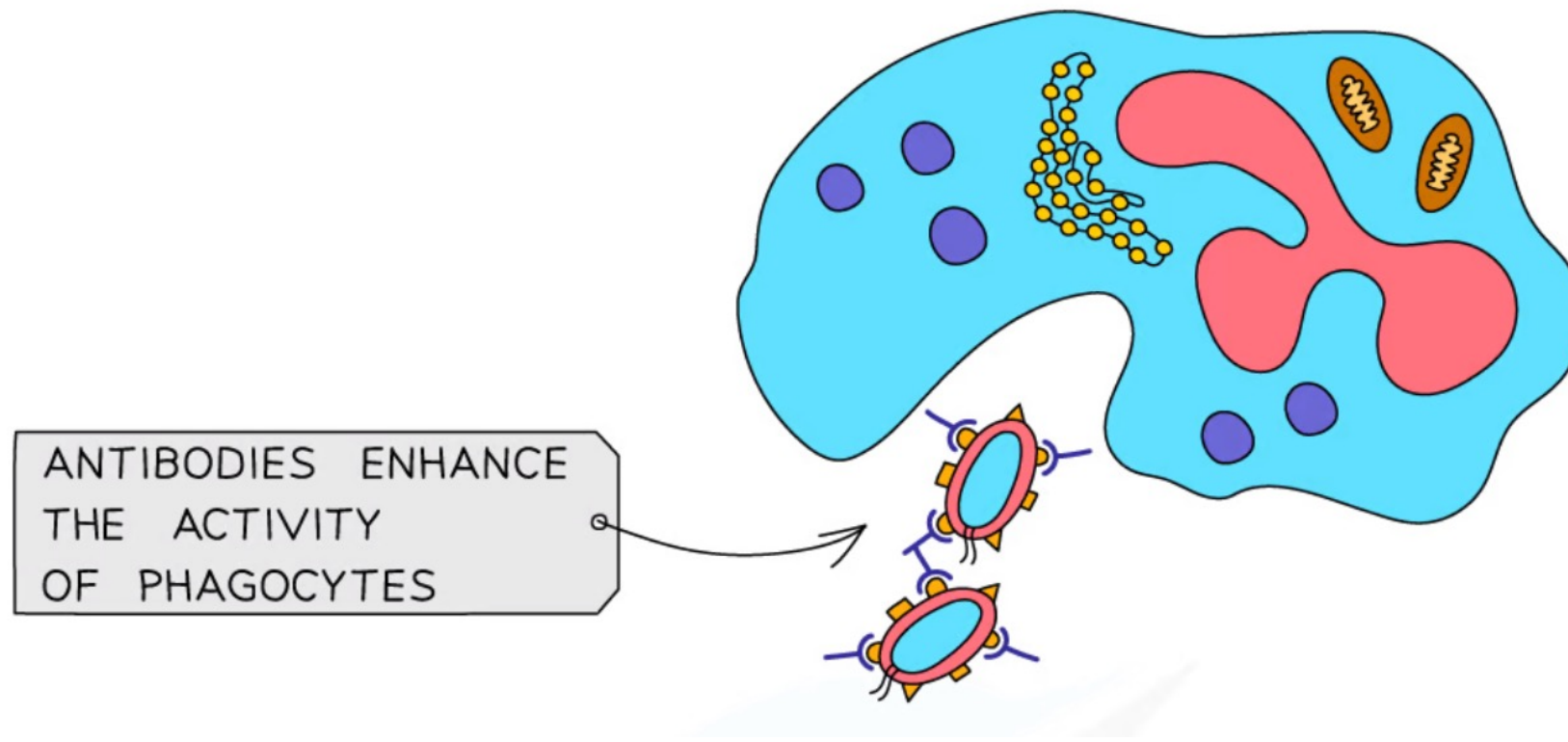


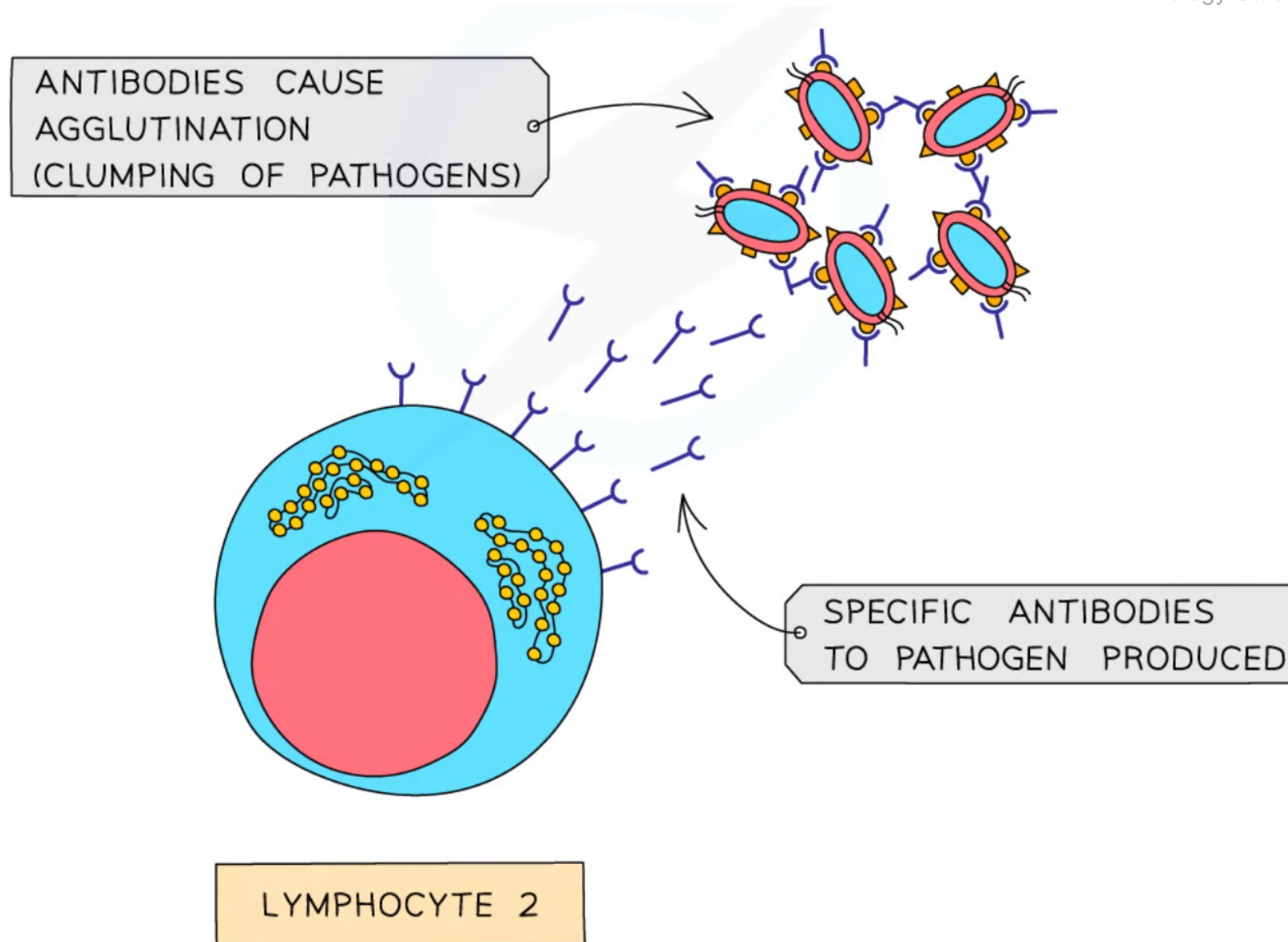
Types of specific immunity





- The antibodies attach to the antigens and cause **agglutination** (clumping together)
- This means the pathogenic cells cannot move very easily
- At the same time, **chemicals** are released that signal to **phagocytes** that there are cells present that need to be destroyed





- The **initial response** of a lymphocyte encountering a pathogen for the first time and making specific antibodies for its antigens can take **a few days**, during which time an individual may get sick
- Lymphocytes that have made antibodies for a specific pathogen for the first time will then make '**memory cells**' that retain the instructions for making those specific antibodies for that type of pathogen
- This means that, in the case of reinfection by the same type of pathogen, **antibodies can very quickly be made** in greater quantities and the pathogens destroyed before they are able to multiply and cause illness
- This is how people can become **immune** to certain diseases after only having them once
- It does not work with all disease-causing microorganisms as some of them **mutate** fairly quickly and change the antigens on their cell surfaces
- Therefore, if they invade the body for a second time, the memory cells made in the first infection will not recall them as they now have slightly different antigens on their surfaces (e.g. the cold virus)





Diseases Caused by the Immune System

- Occasionally, the cells of the immune system start to **attack the body's own cells**
- This is rare as lymphocytes usually **recognise** their own body cells by the **antigens** on the cell surfaces and do not respond to them
- In this situation, **specific body cells are targeted by lymphocytes** and antibodies are made against them, destroying them
- One example of this type of disease is **Type 1 diabetes**
- People who suffer from this disease no longer make their own insulin and so are unable to regulate their blood glucose levels
- This is because their immune system is **targeting and destroying the pancreatic cells which are responsible for making the insulin**, eventually leading to dangerously high glucose levels in the blood



There is often a lot of confusion amongst students regarding the terms antigen, antibody and antibiotic: An **antigen** is a chemical found on the surface of a cell.

An **antibody** is a chemical made by lymphocytes that is complementary to an antigen and, when attached, clumps them together and signals the cells they are on for destruction.

An **antibiotic** is a drug that slows down or stops the growth of bacteria.



- 10 Explain that active immunity is gained after an infection by a pathogen or by vaccination
- 11 Outline the process of vaccination:
 - (a) weakened pathogens or their antigens are put into the body
 - (b) the antigens stimulate an immune response by lymphocytes which produce antibodies
 - (c) memory cells are produced that give long-term immunity
- 12 Explain the role of vaccination in controlling the spread of diseases

S

Vaccination

Vaccination involves receiving a **vaccine** for the body to induce active immunity.

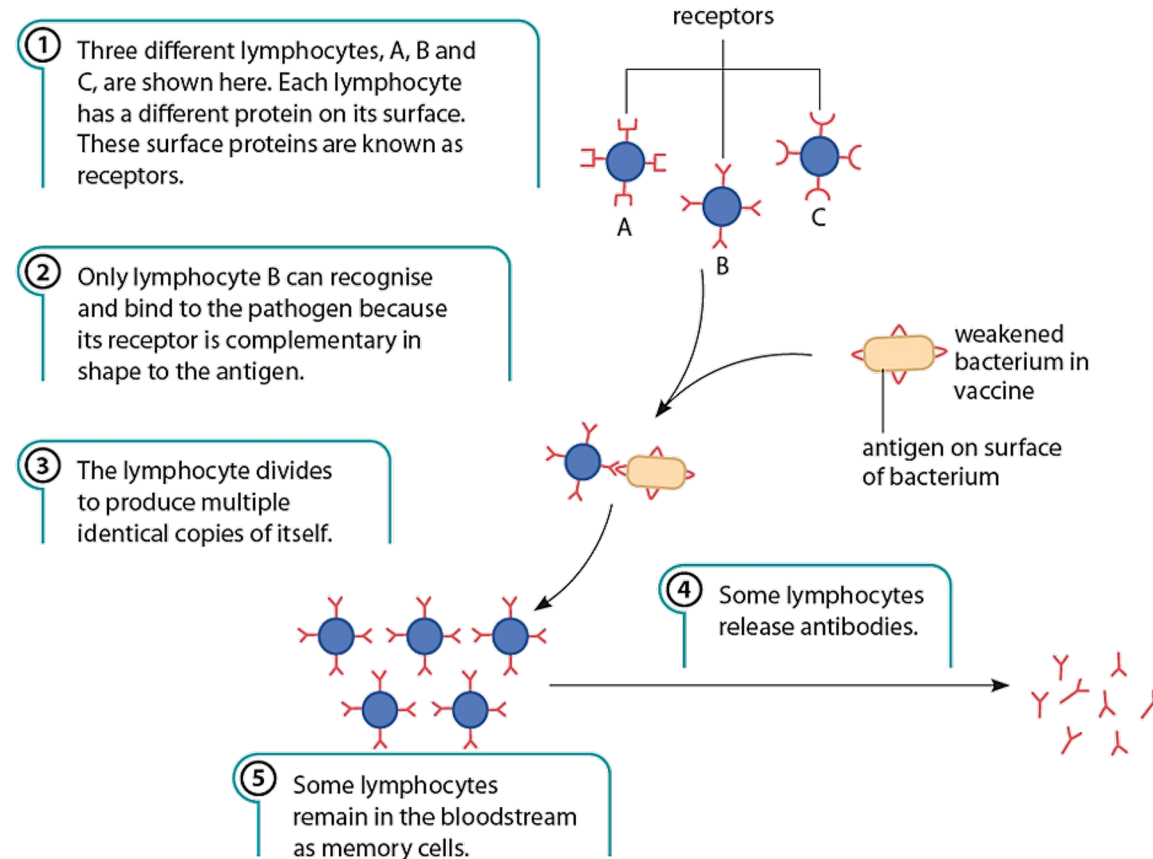
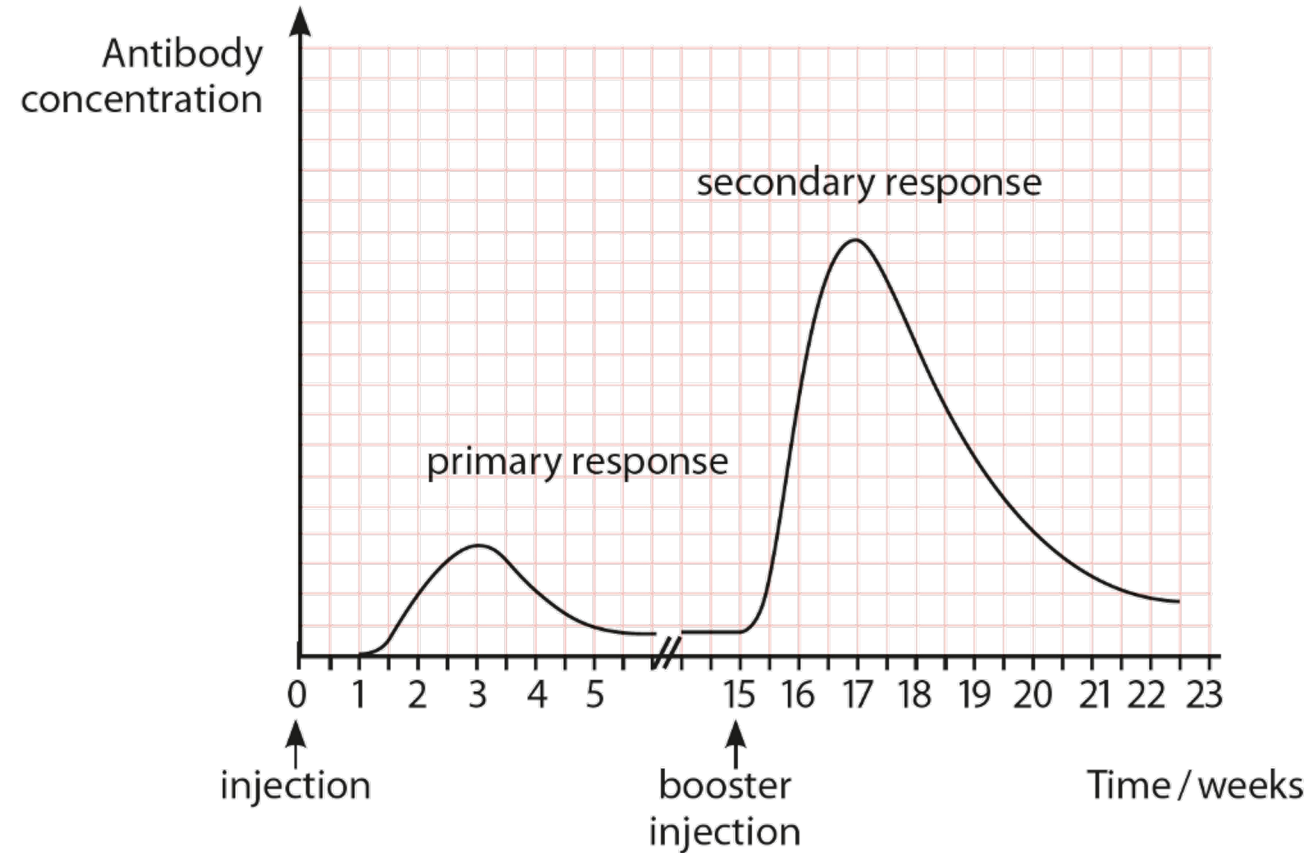


Diagram showing how a vaccine works

Vaccination



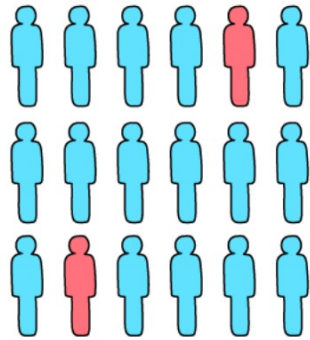
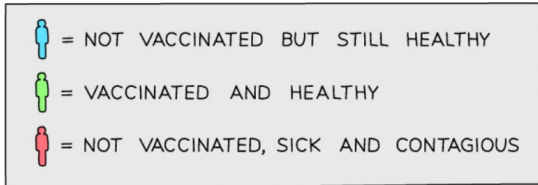
Concentration of antibodies in the blood after the infection of a vaccine and a booster



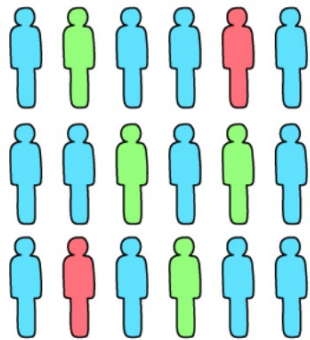
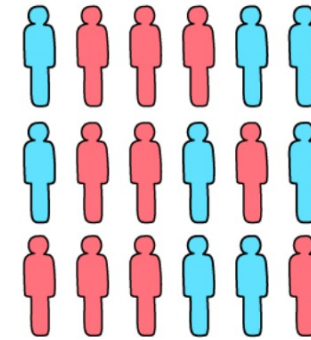
How does Vaccination Control the Spread of Disease

- If a **large enough percentage** of the population is vaccinated, it **provides protection for the entire population** because there are **very few places for the pathogen to breed** - it can only do so if it enters the body of an unvaccinated person
- This is known as **herd immunity**
- If the number of people vaccinated against a specific disease **drops** in a population, it leaves the rest of the population at risk of **mass infection**, as they are more likely to come across people who are infected and contagious

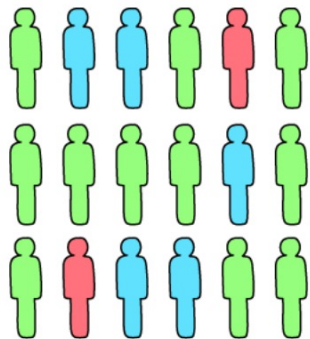
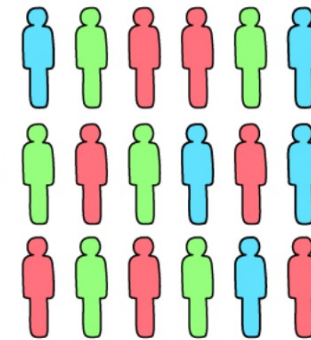
This **increases the number of infections**, as well as the number of people who could die from a specific infectious disease



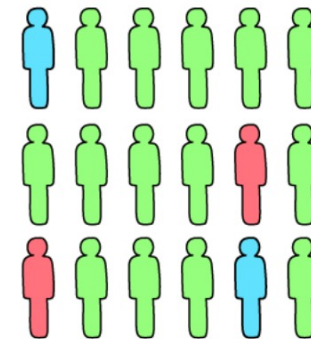
NO ONE IS VACCINATED.
CONTAGIOUS DISEASE
SPREADS THROUGH THE
POPULATION



SOME OF THE POPULATION
GETS VACCINATED.
CONTAGIOUS DISEASE
SPREADS THROUGH SOME
OF THE POPULATION.



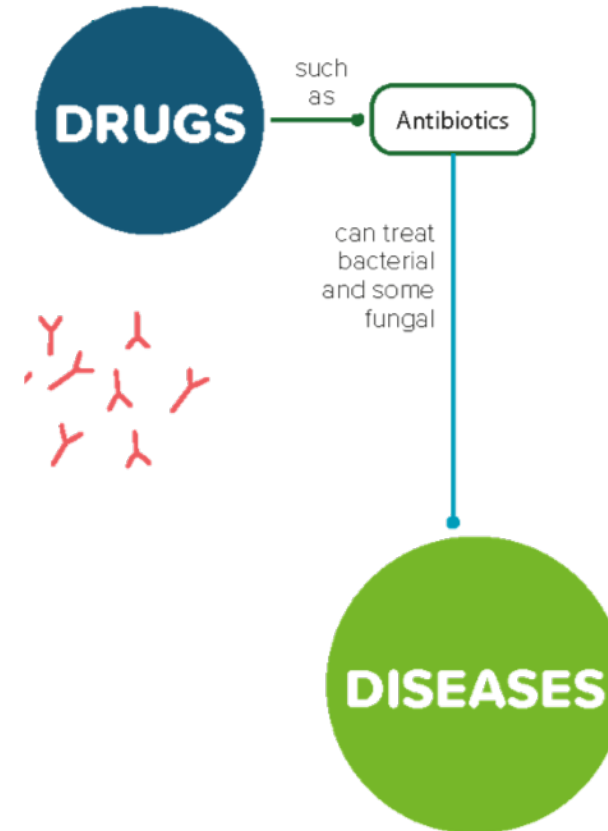
MOST OF THE POPULATION
GETS VACCINATED.
SPREAD OF CONTAGIOUS
DISEASE IS CONTAINED.



10.3 Drugs

In this section, you will learn the following:

- Describe a drug as any substance taken into the body that modifies or affects chemical reactions in the body.
- Describe the use of antibiotics for the treatment of bacterial infection.
- State that antibiotics kill bacteria but do not affect viruses.
- State that some bacteria are resistant to antibiotics, which reduces the effectiveness of antibiotics.
- **S** Explain how using antibiotics only when essential can limit the development of resistant bacteria such as MRSA.





15.1 Drugs

Core

- 1 Describe a drug as any substance taken into the body that modifies or affects chemical reactions in the body
- 2 Describe the use of antibiotics for the treatment of bacterial infections
- 3 State that some bacteria are resistant to antibiotics which reduces the effectiveness of antibiotics
- 4 State that antibiotics kill bacteria but do not affect viruses

Supplement

- 5 Explain how using antibiotics only when essential can limit the development of resistant bacteria such as MRSA

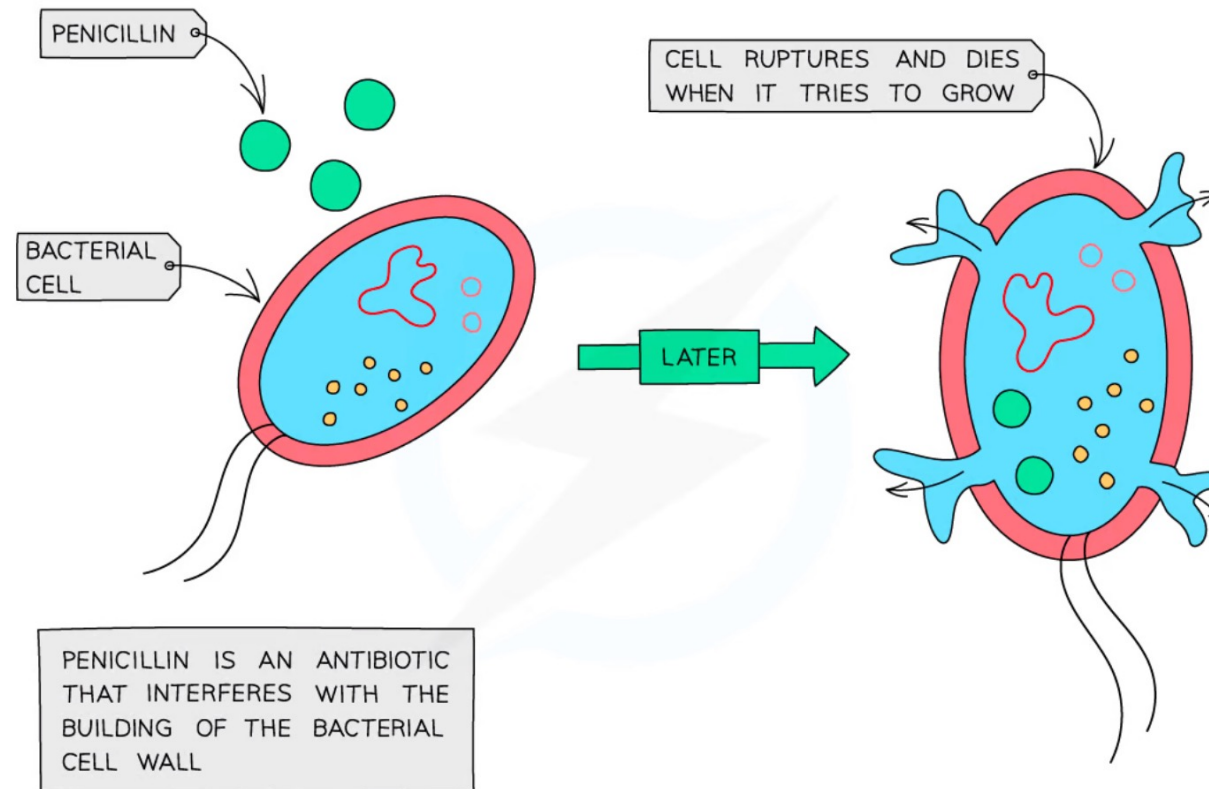


What is a Drug?

- A drug is **any substance taken into the body that modifies or affects chemical reactions in the body**
- Some drugs are medicinal drugs that are used to treat the symptoms or causes of a disease - for example, **antibiotics**

Antibiotics

- Antibiotics are **chemical substances** made by certain **fungi or bacteria** that affect the working of bacterial cells, either by **disrupting their structure or function** or by **preventing them from reproducing**.
- Antibiotics are **effective against bacteria but not against viruses**.
- Antibiotics target processes and structures that are specific to bacterial (prokaryotic) cells; as such they do not generally harm animal cells.



How do antibiotics work?

Antibiotics are drugs used to treat bacterial and some fungal infections. They interfere with some aspects of growth or metabolic activities of the pathogens.

Antibiotics inhibit:

- **Synthesis of bacterial cell wall**
- **Protein synthesis**
- **Enzyme action**
- **Cell membrane**



Antibiotics

WORD ALERT



Inhibits: hinders or prevents



Effectiveness of Antibiotics

- Some bacteria that cause disease have become **resistant** to antibiotics and this **reduces the effectiveness of prescribed antibiotics** when someone has a bacterial infection, as it might be caused by a type of bacteria that is resistant to that particular antibiotic

Why don't Antibiotics Affect Viruses?

- Viruses cannot be treated with antibiotics
- This is because antibiotics work by **disrupting cell functions such as respiration, or breaking down the structure of the cell in some way**
- However, viruses **do not carry out any cell functions** and **do not have cell walls, cell membranes or any cell organelles** as viruses infect and utilise the machinery of animal cells to reproduce, which are not affected by antibiotics.
- Therefore the action of antibiotics **do not affect them**



Why do antibiotics kill bacteria but not viruses?

Differences between viruses and bacteria

Viruses	Bacteria
Have a protein coat	Have a cell wall
Cell membrane absent	Cell membrane present
Cytoplasm absent	Cytoplasm present
Ribosomes absent	Ribosomes present

Antibiotics does not affect viruses because viruses do not have structures that can be targeted.

How do bacteria develop resistance to antibiotics?

When bacteria are exposed to certain chemicals or agents, their genes might change. This is known as **mutation**.

Some mutations allow bacteria to become resistant to antibiotics. When these bacteria reproduce, the mutation will be passed on to future generations.



S

Why do doctors always tell their patients to complete the full course of antibiotics?

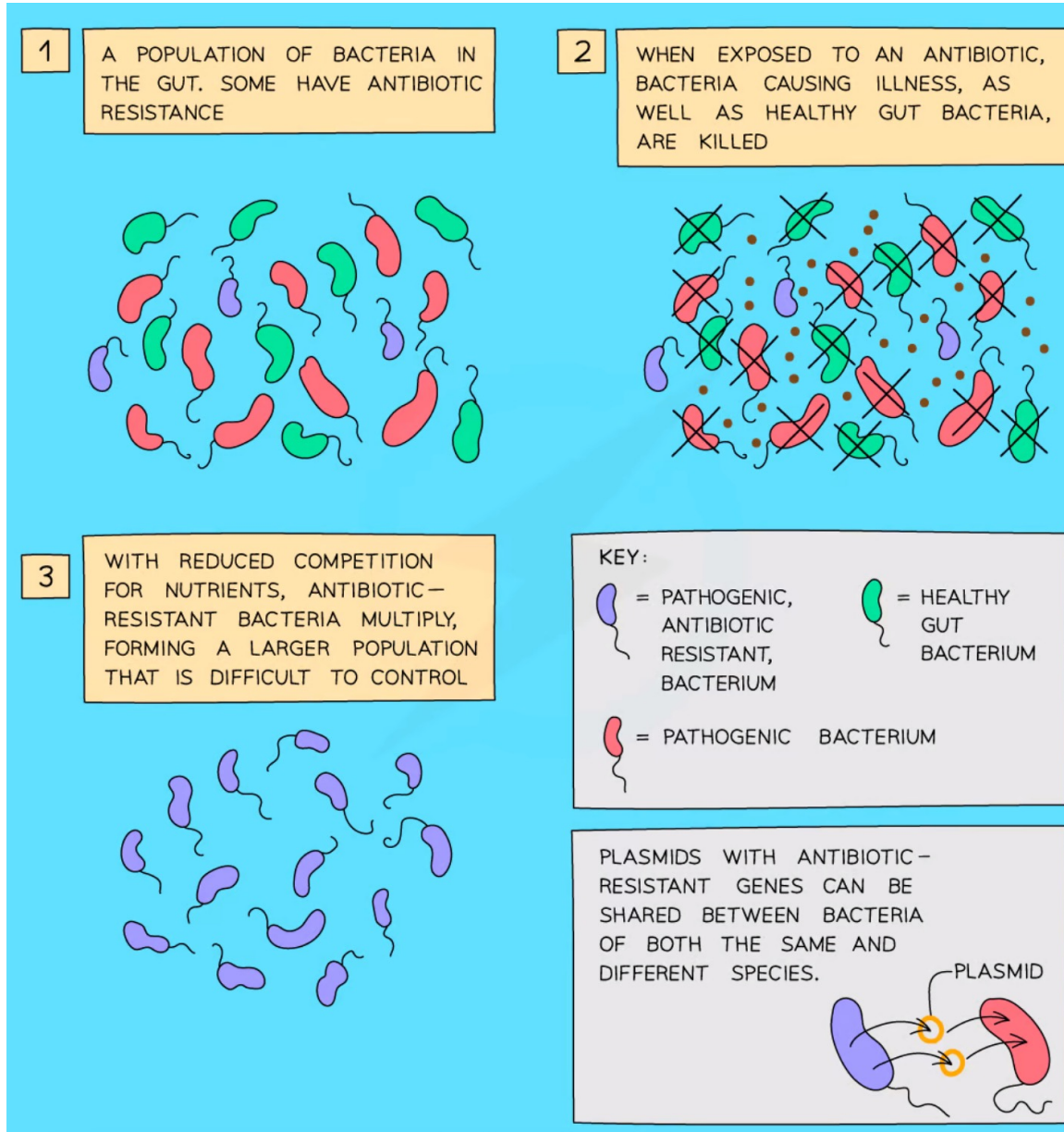
Taking the full course of antibiotics ensures that the infection is properly treated. This prevents the survival of bacteria that may become antibiotic-resistant.





Antibiotic Resistance

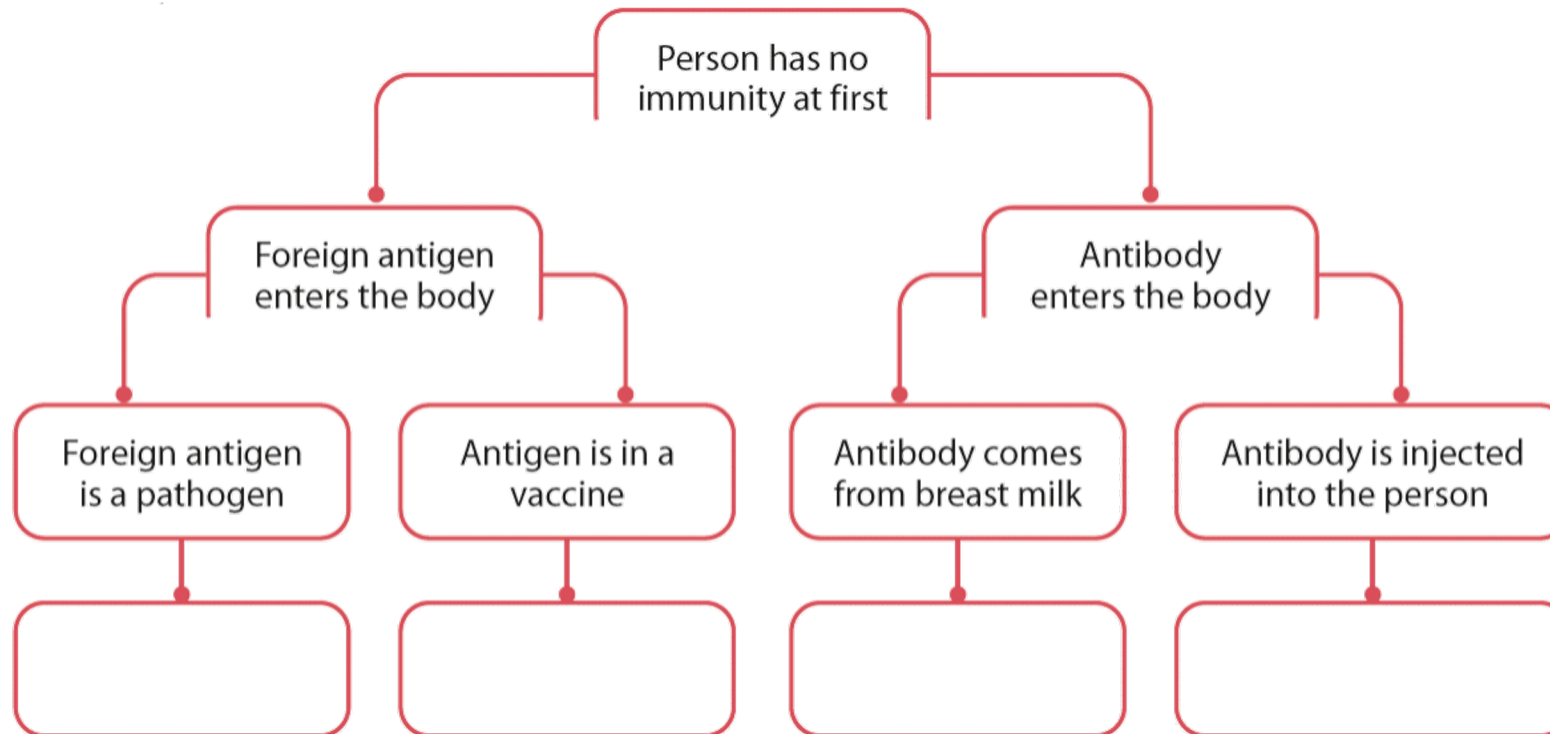
- Since the first antibiotic was discovered in 1928, many more have been discovered and developed and antibiotics were and are **widely overused**
- Commonly prescribed antibiotics are becoming **less effective** due to a number of reasons:
 - **overuse** and being prescribed when not really necessary
 - patients **failing to complete the fully prescribed course** by a doctor
 - large scale **use of antibiotics in farming** to prevent disease when livestock are kept in close quarters, even when animals are not actually sick
- This has led to the effectiveness of antibiotics being reduced, and the incidence of antibiotic resistance increasing
- These bacteria are commonly known as superbugs and the most common is **MRSA**
- Ways individuals can help **prevent** the incidence of antibiotic resistance increasing include:
 - only taking antibiotics when **absolutely essential**
 - when prescribed a course of antibiotics, **ensure that the entire course is completed** even if you feel better after a few days





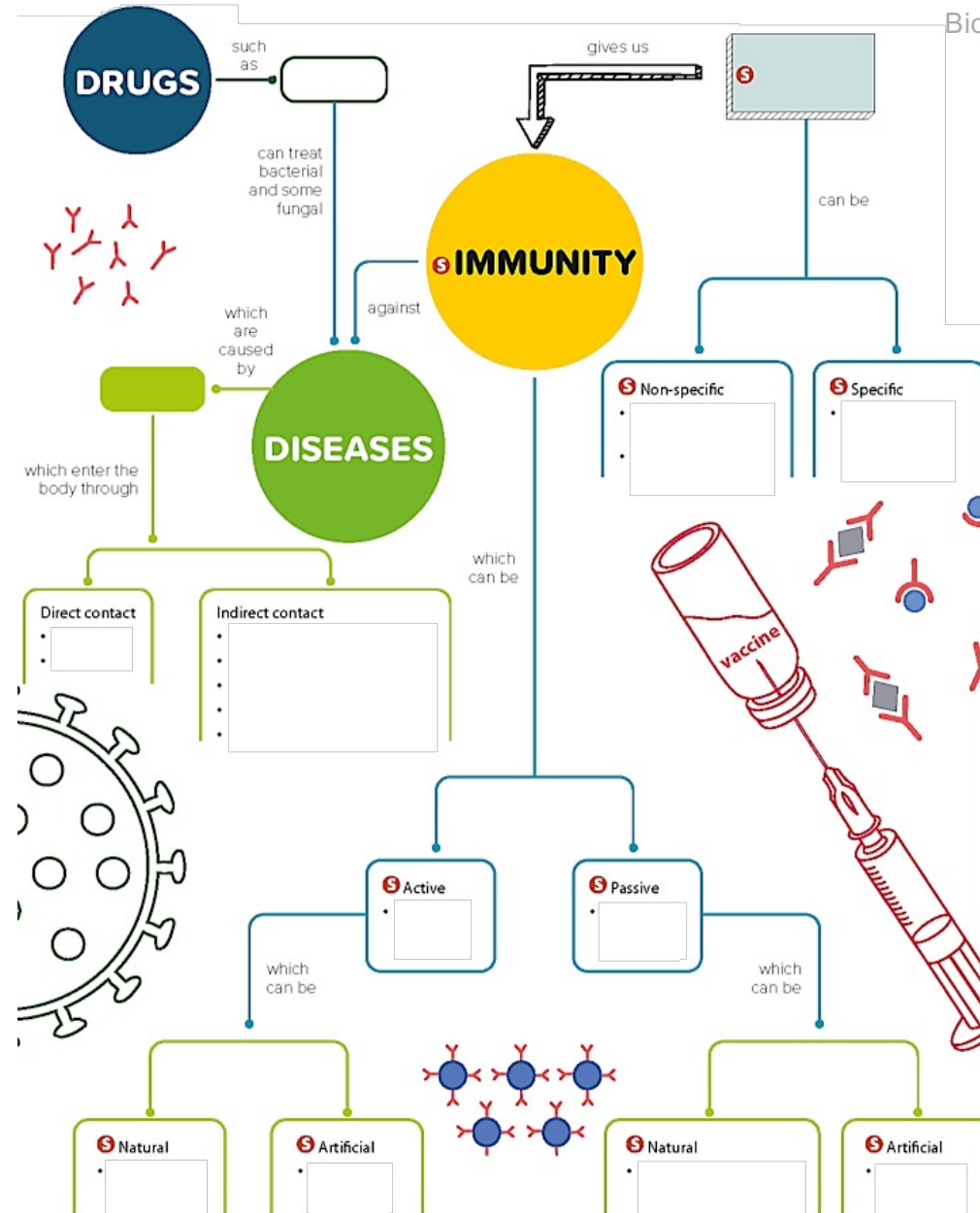
Let's Practise 10.2 and 10.3

- 1 **S** Complete the flowchart in the figure below by writing the correct type of immunity in each empty box.



What have you learned?

Can you draw your own mind map?



What have you learned?

Can you draw your own mind map?

